$$\frac{\partial^2 \hat{U}'\hat{U}}{\partial\hat{\beta}\ \partial\hat{\beta}'} = 2\ \boldsymbol{X'X} \qquad \dots (6.16)$$

Since $\boldsymbol{X'X}$ is a positive definite matrix, we find that $\widehat{\boldsymbol{\beta}}$ represents the minimum of the RSS.

The estimated regression model is given by

$$\widehat{Y} = \boldsymbol{X}\widehat{\boldsymbol{\beta}} \qquad \dots (6.17)$$

6.3.1 Mean of OLS Estimators

From equation (6.15) we know that

$\widehat{\boldsymbol{\beta}} = (\boldsymbol{X'X})^{-1}\ \boldsymbol{X'Y}$. By substituting the value of Y from equation (6.1), we obtain

$$\widehat{\boldsymbol{\beta}} = (\boldsymbol{X'X})^{-1}\ \boldsymbol{X'}(\boldsymbol{X\beta} + \boldsymbol{U}) = (\boldsymbol{X'X})^{-1}\ \boldsymbol{X'X\beta} + (\boldsymbol{X'X})^{-1}\ \boldsymbol{X'U}$$

Thus,

$$\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\boldsymbol{X'X})^{-1}\boldsymbol{X'U} \qquad \dots (6.18)$$

Mean of $\widehat{\boldsymbol{\beta}} = E(\widehat{\boldsymbol{\beta}}) = E[\boldsymbol{\beta} + (\boldsymbol{X'X})^{-1}\boldsymbol{X'U}]$

Since $\boldsymbol{X}$ is kept fixed in repeated samples, it is not stochastic or random. Further, by assumption $E(\boldsymbol{U}) = \boldsymbol{0}$

Therefore,

$$E(\widehat{\boldsymbol{\beta}}) = \boldsymbol{\beta} + (\boldsymbol{X'X})^{-1}\boldsymbol{X'}E(\boldsymbol{U}) = \boldsymbol{\beta} \qquad \dots (6.19)$$

The expected value of the OLS estimator $\widehat{\boldsymbol{\beta}}$ is the true parameter $\boldsymbol{\beta}$. An implication of the above is that the OLS estimators are *unbiased*.

6.3.2 Variance of OLS Estimators

Now let us find out the variance of the OLS estimator $\widehat{\boldsymbol{\beta}}$. Recall that $\widehat{\boldsymbol{\beta}}$ is a $k \times 1$ vector of the estimates. The values of the estimates, vary across samples. The variance of the OLS estimators $\widehat{\boldsymbol{\beta}}$ is given by

$$var(\widehat{\boldsymbol{\beta}}) = E\left[(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta})(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta})'\right] \qquad \dots (6.20)$$

In fact, (6.19) represents the variance-covariance matrix of the parameters in the regression model. It is a matrix of the order $(k \times k)$. The diagonal terms of this matrix represent the variance of the parameters while the off-diagonal terms represent the covariance between the parameters (for example, the ij[th] term of the matrix $\left[(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta})(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta})'\right]$ represents the covariance between β_i and β_j.

From (6.18), we find that $\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\boldsymbol{X'X})^{-1}\boldsymbol{X'U}$

Therefore,

$$E\left[(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta})(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta})'\right] = E\left[(\widehat{\boldsymbol{X}} - \boldsymbol{X})^{-1}\boldsymbol{X'UU'X}\ (\boldsymbol{X'X})^{-1}\right]$$

$$= (\boldsymbol{X'} - \boldsymbol{X})^{-1}\boldsymbol{X'}\ \boldsymbol{E}\ (\boldsymbol{UU'})\ \boldsymbol{X}\ (\boldsymbol{X'X})^{-1}$$

Since $\boldsymbol{E}\ (\boldsymbol{UU'}) == \sigma^2\boldsymbol{I}$ by assumption, we have

$$var(\hat{\boldsymbol{\beta}}) = \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1} \quad \dots (6.21)$$

Note that σ^2 is a constant and the order of the symmetric matrix $(\boldsymbol{X}'\boldsymbol{X})$is k.

6.3.3 Estimator of Error Variance

For finding the variance of $\hat{\beta}$ we need an estimator of σ^2. As you know, the error variable cannot be observed. We take the residuals as a proxy of the error variable. Let us find out the variance of the stochastic error in terms of the residuals.

In Unit 4, while dealing with two-variable regression models, we found that an unbiased estimator of the error variance is given by $RSS/(n-2)$. Recall that there are two parameters (intercept and slope) in the two-variable regression model. We will generalise the above result to cases where the number of explanatory variables is k.

We know that the residual is given by $\hat{\boldsymbol{U}} = (\boldsymbol{Y} - \hat{\boldsymbol{Y}})$. Let us substitute the value of $\hat{\boldsymbol{Y}}$ from equation (6.17).

$$\hat{\boldsymbol{U}} = \boldsymbol{Y} - \boldsymbol{X}\hat{\boldsymbol{\beta}} = \boldsymbol{Y} - \boldsymbol{X}\,(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Y}$$

$$= [\boldsymbol{I} - \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}']\boldsymbol{Y}$$

$$= \boldsymbol{MY} \quad \dots (6.22)$$

where $\boldsymbol{M} = [\boldsymbol{I} - \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}']$

Remember that $\boldsymbol{M}$ is an 'idempotent matrix'. Thus, $\boldsymbol{M}$ is a symmetric matrix of order *n*. Further, if we multiply $\boldsymbol{M}$ with itself, its value remains unchanged. Thus,

$$\boldsymbol{M} = \boldsymbol{M}'$$

and

$$\boldsymbol{MM} = [\boldsymbol{I} - \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'][\boldsymbol{I} - \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}']$$

$$= [\boldsymbol{I} - \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' - \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' + \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}']$$

$$= \boldsymbol{I} - \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' - \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' + \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$$

$$= \boldsymbol{I} - \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' = \boldsymbol{M}$$

Further,

$$\mathbf{MX} = [\boldsymbol{I} - \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}']\boldsymbol{X} = \boldsymbol{X} - \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{X} = \boldsymbol{X} - \boldsymbol{X} = \mathbf{0} \quad \dots (6.23)$$

From (6.22) we have $\hat{\boldsymbol{U}} = \boldsymbol{MY}$.We can write$\boldsymbol{MY}$ as

$$\boldsymbol{MY} = \boldsymbol{M}(\boldsymbol{X\beta} + \boldsymbol{U}) = \boldsymbol{MX\beta} + \boldsymbol{MU} = \mathbf{0} + \boldsymbol{MU} = \boldsymbol{MU} \quad \dots (6.24)$$

The **RSS** is given by$\sum \hat{u}_i^2$ which can be written in matrix form as$\hat{\boldsymbol{U}}'\hat{\boldsymbol{U}}$. You should note the difference between the matrix $\boldsymbol{U}'\boldsymbol{U}$and the matrix $\boldsymbol{UU}'$. The matrix $\boldsymbol{UU}'$ (of the order $(n \times n)$) represents the variance-covariance of the error term. On the other hand, matrix $\boldsymbol{U}'\boldsymbol{U}$(of the order (1×1)) is a scalar quantity.

$$\hat{\boldsymbol{U}}'\hat{\boldsymbol{U}} = (\boldsymbol{MU})'(\boldsymbol{MU}) = \boldsymbol{U}'\boldsymbol{M}'\boldsymbol{MU} = \boldsymbol{U}'\boldsymbol{MU} \quad \dots (6.25)$$

Note that $\boldsymbol{U'MU}$ is a scalar as the order of the matrix is $(1 \times n)(n \times n)(n \times 1) = (1 \times 1)$. Further, recall the concept of trace of a matrix discussed in Unit 2. Trace of a matrix is the sum of the diagonal elements of the matrix. Thus, trace of a scalar is the scalar itself. Let us point out two properties of the trace of a square matrix.

(a) For matrix **A**, we have $Tr\,(\boldsymbol{A}) = \sum a_{ii}$
(b) For two matrices **A** and **B**, we have $Tr\,(\boldsymbol{AB}) = Tr\,(\boldsymbol{BA})$

By taking expected value of (6.23), we obtain

$$E\left(\boldsymbol{\hat{U}'\hat{U}}\right) = E(\boldsymbol{U'MU}) = E[Tr(\boldsymbol{U'MU})] = E[Tr(\boldsymbol{U'MU})]$$

$$E(\boldsymbol{U'MU}) = E[Tr(\boldsymbol{U'MU})] = E[Tr(\boldsymbol{UU'M})]$$

(Assume that $\boldsymbol{U'M} = \boldsymbol{A}$ and $\boldsymbol{U} = \boldsymbol{B}$)

Thus,

$$E[Tr(\boldsymbol{UU'M})] = \sigma^2\,Tr(\boldsymbol{M})$$
$$= \sigma^2\,Tr[\boldsymbol{I} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'}] = \sigma^2\,Tr(\boldsymbol{I}) - \sigma^2 Tr[\boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'}]$$

$$= \sigma^2\,Tr(\boldsymbol{I}) - \sigma^2 Tr[\boldsymbol{X'X}(\boldsymbol{X'X})^{-1}]$$

$$= \sigma^2\,n - \sigma^2 k$$

$$= \sigma^2\,(n-k)$$

Thus,
$$E\left(\boldsymbol{\hat{U}'\hat{U}}\right) = \sigma^2\,(n-k)$$

By re-arranging terms, we obtain an estimator of σ^2. Let us call it $\hat{\sigma}^2$.

$$\hat{\sigma}^2 = E\left(\boldsymbol{\hat{U}'\hat{U}}\right)/(n-k) = \boldsymbol{RSS}/(n-k) \qquad \ldots (6.26)$$

Check Your Progress 1

1) Bring out the advantages of multiple regression models over bivariate regression models.

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2) Derive the expression for the OLS estimator in the following linear regression model:

$\boldsymbol{Y = X\beta + U.}$

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3) Derive the expression for the variance of the OLS estimator in the following regression model; $\boldsymbol{Y} = \boldsymbol{X\beta} + \boldsymbol{U}$.

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4) Why is it important that the $\boldsymbol{X}$ matrix should have full rank in the regression model; $\boldsymbol{Y} = \boldsymbol{X\beta} + \boldsymbol{U}$?

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6.4 PROPERTIES OF OLS ESTIMATORS

A good estimator should satisfy certain statistical properties such as unbiasedness, consistency and efficiency. In this section we will examine whether the OLS estimator fulfils these properties. In addition, there are certain algebraic properties of the OLS estimators. We begin with the algebraic properties.

6.4.1 Algebraic Properties

(i) The normal equations of the regression model are given by

$$(\boldsymbol{X'X})\widehat{\boldsymbol{\beta}} - \boldsymbol{X'Y} = \boldsymbol{0}$$

Let us substitute the value of $\boldsymbol{Y}$ from equation (6.1) in the above equation, we get

$$(\boldsymbol{X'X})\widehat{\boldsymbol{\beta}} - \boldsymbol{X'}\left(\boldsymbol{X}\widehat{\boldsymbol{\beta}} + \widehat{\boldsymbol{U}}\right) = \boldsymbol{0}$$

Or, $(\boldsymbol{X'X})\widehat{\boldsymbol{\beta}} - (\boldsymbol{X'X})\widehat{\boldsymbol{\beta}} + \boldsymbol{X'}\widehat{\boldsymbol{U}} = \boldsymbol{0}$

Or,

$$\boldsymbol{X'}\widehat{\boldsymbol{U}} = \boldsymbol{0} \quad \ldots (6.27)$$

Equation (6.25) implies that the residual and the explanatory variables are not correlated.

(ii) The predicted value $\widehat{\boldsymbol{Y}}$ and the residual $\widehat{\boldsymbol{U}}$ are not correlated.

$$\widehat{\boldsymbol{Y}}'\widehat{\boldsymbol{U}} = (\boldsymbol{X}\widehat{\boldsymbol{\beta}})'\widehat{\boldsymbol{U}} = \widehat{\boldsymbol{\beta}}'\boldsymbol{X'}\widehat{\boldsymbol{U}} = \boldsymbol{0} \quad \ldots (6.28)$$

6.4.2 Statistical Properties

Let us discuss about the statistical or theoretical properties of OLS estimators.

(a) **Unbiasedness:** In the regression equation, we have added an error variable (**U**), which is random. The addition of a random variable to the regression model makes the dependent variable (**Y**) also random. Since the estimator ($\widehat{\boldsymbol{\beta}}$) is a linear combination of the dependent variable, the estimator will provide estimates which are random. Unbiasedness property of an estimator can be checked by taking repeated samples. It says that the average of the estimates will be equal to the parameters if we take repeated samples. In technical terms, the expected value of the estimator is equal to the parameter, i.e., $E(\widehat{\boldsymbol{\beta}}) = \boldsymbol{\beta}.$ This we have proved in equation (6.18). Thus, the OLS estimators are unbiased.

(b) **Consistency**: The consistency property of an estimator is different from unbiasedness. An estimator is said to be consistent if its value approaches the parameter value as the sample size is increased. We know from equation (6.17) that $\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{U}.$ To prove that $\widehat{\boldsymbol{\beta}}$is consistent, we have to prove that the second term on the right-hand side has a probability limit (*plim*) of zero. This term is the product of two terms $(\boldsymbol{X}'\boldsymbol{X})^{-1}$and $\boldsymbol{X}'\boldsymbol{U}$. Neither $(\boldsymbol{X}'\boldsymbol{X})^{-1}$nor $\boldsymbol{X}'\boldsymbol{U}$ has a probability limit. However, $(n^{-1} \times n) = 1$. We can write $(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{U}$ as $\left(\frac{\mathbf{1}}{\boldsymbol{n}}\boldsymbol{X}'\boldsymbol{X}\right)^{-\mathbf{1}}\frac{\mathbf{1}}{\boldsymbol{n}}\boldsymbol{X}'\boldsymbol{U}.$ Therefore,

$$plim(\widehat{\boldsymbol{\beta}}) = \boldsymbol{\beta} + \left(plim\frac{\mathbf{1}}{\boldsymbol{n}}\boldsymbol{X}'\boldsymbol{X}\right)^{-\mathbf{1}} plim\frac{\mathbf{1}}{\boldsymbol{n}}\boldsymbol{X}'\boldsymbol{U} = \boldsymbol{\beta} \quad \ldots (6.29)$$

(c) **Efficiency**: efficiency of an estimator refers to the 'minimum variance unbiased estimator'. Since the variance of the estimator is the lowest, the estimates will be scattered closer to the parameter. Consequently, the estimator will give better and more accurate estimates than other estimators having higher variance. Notice, the comparison is among unbiased estimators only. As we will see below, the OLS estimator is the 'best linear unbiased estimator' (BLUE).

6.5 BEST LINEAR UNBIASED ESTIMATOR (BLUE)

We are given a sample data set, which contains data on ***X*** and ***Y***. We maintain the classical assumptions about the stochastic error term $\boldsymbol{U}.$ We have two estimators: the OLS estimator $\widehat{\boldsymbol{\beta}}$, and any other linear unbiased estimator $\widetilde{\boldsymbol{\beta}}.$ We have to prove that the OLS estimator $\widehat{\boldsymbol{\beta}}$ is the best.

This property of the OLS estimator is also called the 'Gauss-Markov theorem'. The OLS estimator is given by

$$\widehat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Y} \quad \ldots (6.14)$$

Here, the ***X*** matrix is assumed to be kept fixed in repeated samples. Further, we know the value of **X** from the sample of data available to us. We can write $\widehat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Y} = \boldsymbol{W}\boldsymbol{Y}$where $\boldsymbol{W} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$. Thus,$\widehat{\boldsymbol{\beta}}$ is a linear combination of ***Y*** and $\widehat{\boldsymbol{\beta}}$is a linear estimator.

In order to prove that $\widehat{\boldsymbol{\beta}}$is BLUE, we compare it with another *linear unbiased* estimator, say, $\widetilde{\boldsymbol{\beta}}$. Since $\widetilde{\boldsymbol{\beta}}$is a linear estimator, we can consider

$$\widetilde{\boldsymbol{\beta}} = \boldsymbol{C}\boldsymbol{Y} \quad \ldots (6.30)$$

By substituting the value of ***Y*** from the regression model (6.1), we can write (6.30) as

$$\widetilde{\boldsymbol{\beta}} = \mathbf{C}\,(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{U}) = \boldsymbol{C}\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{C}\boldsymbol{U} \quad \ldots (6.31)$$

Note that we are comparing amongst unbiased estimators. Thus, we should have

$E(\widetilde{\boldsymbol{\beta}}) = \boldsymbol{\beta}$ … (6.32)

By using the value of $\widetilde{\boldsymbol{\beta}}$ from equation (6.30), we obtain

$E(\widetilde{\boldsymbol{\beta}}) = E(\boldsymbol{CX\beta} + \boldsymbol{CU}) = \boldsymbol{CX\beta} + \boldsymbol{C}E(\boldsymbol{U})$ … (6.33)

From the above, $E(\widetilde{\boldsymbol{\beta}}) = \boldsymbol{\beta}$ (i.e., $\widetilde{\boldsymbol{\beta}}$ is unbiased) only if we assume that

$\boldsymbol{CX} = \boldsymbol{I}$ … (6.34)

Let us find out the variance of $\widetilde{\boldsymbol{\beta}}$. We can re-write equation (6.31) as

$\widetilde{\boldsymbol{\beta}} - \boldsymbol{\beta} = \boldsymbol{CU}$ … (6.35)

The variance of $\widetilde{\boldsymbol{\beta}}$ is given by

$var(\widetilde{\boldsymbol{\beta}}) = E\left[(\widetilde{\boldsymbol{\beta}} - \boldsymbol{\beta})(\widetilde{\boldsymbol{\beta}} - \boldsymbol{\beta})'\right]$ … (6.36)

By using (6.34), we can re-write (6.35) as

$var(\widetilde{\boldsymbol{\beta}}) = E[(\boldsymbol{CUU'C'})] = \boldsymbol{C}\,E(\boldsymbol{UU'})\boldsymbol{C'} = \sigma^2\boldsymbol{CC'}$ … (6.37)

The variance of the OLS estimator $var(\widehat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X'X})^{-1}$ whereas the variance of any other linear unbiased estimator $var(\widetilde{\boldsymbol{\beta}}) = \sigma^2\boldsymbol{CC'}$. Since $\widehat{\boldsymbol{\beta}}$ and $\widetilde{\boldsymbol{\beta}}$ are linear estimators, let

$\mathbf{C} = (\mathbf{X'X})^{-1}\mathbf{X'} + \mathbf{D}$ … (6.38)

where $\mathbf{D}$ is a $(k \times n)$ matrix (i.e., the same order as the $(\mathbf{X'X})^{-1}\mathbf{X'}$ matrix).

Let us multiply equation (6.37) by $\boldsymbol{X}$.

$\boldsymbol{CX} = [(\boldsymbol{X'X})^{-1}\boldsymbol{X'} + \boldsymbol{D}]\boldsymbol{X} = (\boldsymbol{X'X})^{-1}\boldsymbol{X'X} + \boldsymbol{DX} = \boldsymbol{I} + \boldsymbol{DX}$

From equation (6.33), we know that $\boldsymbol{CX} = \boldsymbol{I}$. Therefore,

$\boldsymbol{DX} = \mathbf{0}$ … (6.39)

Remember that $var(\widehat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X'X})^{-1}$ and $var(\widetilde{\boldsymbol{\beta}}) = \sigma^2\boldsymbol{CC'}$. Let us compare between $\boldsymbol{CC'}$ and $(\mathbf{X'X})^{-1}$.

$$\begin{aligned}
\boldsymbol{CC'} &= [(\boldsymbol{X'X})^{-1}\boldsymbol{X'} + \boldsymbol{D}][(\boldsymbol{X'X})^{-1}\boldsymbol{X'} + \boldsymbol{D}]' \\
&= [(\boldsymbol{X'X})^{-1}\boldsymbol{X'} + \boldsymbol{D}][\boldsymbol{X}\,(\boldsymbol{X'X})^{-1} + \boldsymbol{D'}] \\
&= (\boldsymbol{X'X})^{-1}(\boldsymbol{X'X})(\boldsymbol{X'X})^{-1} + \boldsymbol{DX}\,(\boldsymbol{X'X})^{-1} + (\boldsymbol{X'X})^{-1}\boldsymbol{X'D'} + \boldsymbol{DD'}
\end{aligned}$$

Since $\boldsymbol{DX} = \boldsymbol{X'D'} = \mathbf{0}$, we obtain

$\boldsymbol{CC'} = (\boldsymbol{X'X})^{-1} + \boldsymbol{DD'}$ … (6.40)

Since $\boldsymbol{DD'}$ is positive semi-definite,

$$\boldsymbol{CC'} \geq (\boldsymbol{X'X})^{-1} + \boldsymbol{DD'} \quad \ldots (6.41)$$

Thus, $var(\widehat{\boldsymbol{\beta}})$ is less than $var(\widetilde{\boldsymbol{\beta}})$**.** In other words, the OLS estimators are the best linear unbiased estimators (BLUE).

Check Your Progress 2

1) Prove that the residual and the dependent variable are not correlated in the case of the multiple regression models.

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2) Prove that the OLS estimators are the best linear unbiased estimators (BLUE).

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3) Show that the OLS estimators are unbiased, consistent and efficient.

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6.6 LET US SUM UP

Multiple regression models enable us to include several explanatory variables in the regression model. Further, it allows specification of a general functional form on the relationship between dependent variable and independent variables. Because of its benefits, the multiple regression models are widely used.

In this Unit we specified the multiple regression model in its matrix form. We applied the method of ordinary least squares (OLS) and derived the estimators of the model. The desirable properties of the estimators such as unbiasedness, consistency and efficiency are discussed. The OLS estimators $\widehat{\boldsymbol{\beta}}$ are the best linear unbiased estimator (BLUE) of $\boldsymbol{\beta}$**.** f

6.7 KEY WORDS

Asymptotic Properties	:	It refers to the properties of estimators and test statistics that apply when the sample size grows without bound.

Biased Estimator : An estimator whose expectation, or sample mean, is different from the population value it is supposed to be estimating.

Ceteris Paribus : It refers to the assumption that all other relevant factors are held fixed.

Maximum Likelihood Estimation (MLE) : A broadly applicable estimation method where the estimates are chosen to maximize the log-likelihood function.

Quadratic Functions : It refers to the functions that contain squares of one or more explanatory variables; they capture diminishing or increasing effects on the dependent variable.

6.8 SOME USEFUL BOOKS

Greene, W. H., 2018, "*Econometric Analysis.*" Eighth Edition, Pearson Education

Johnston, J., Dinardo, J., 2007, "*Econometric Methods*", 4th Edition, McGraw-Hill Education (Asia), International Edition, pp. 109-125

Maddala, G. S. and K. Lahiri, 2012, *Introduction to Econometrics*, Fourth Edition, Wiley

Verbeek, M. 2012, "*A Guide to Modern Econometrics*", Fourth Edition, John Wiley & Sons

6.9 ANSWERS/ HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) The advantages of using multiple regression model are as follows:

   a. It enables us to control for several other factors simultaneously which could have influence on the dependent variable.

   b. It helps us examine the effects of an independent variable on the dependent variable ceteris paribus.

   c. It eliminates the chances of bias arising due to correlation between the error term and the explanatory variable which is so peculiar to two-variable models.

   d. It presents a general functional form on the cause and effect relationships between variables.

2) Refer to Section 6.3 and answer.

3) The variance of the OLS estimators is given by $var(\hat{\boldsymbol{\beta}}) = \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-\mathbf{1}}$**.** Please see Sub-Section 6.3.2 for derivation.

4) The OLS estimator is given by $\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Y}$. Full rank of the $\boldsymbol{X}$ matrix implies that the explanatory variables are independent of one another. If any of the columns in the matrix is a linear combination of other columns, we cannot find the matrix $(\boldsymbol{X}'\boldsymbol{X})^{-1}$.

Check Your Progress 2

1) Go through Sub-Section 6.4.1 and answer.
2) Go through Section 6.5 and answer.
3) You have to show that the OLS estimators are unbiased, they are consistent, and they are the best linear unbiased estimator. Go through Section 6.4 and Section 6.5.